

# Cutting-Edge Sensor Technologies

Whilst incremental changes are to be seen everywhere, we bring to you the most cutting-edge technologies in the world of sensors

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It is the beginning of a new year, and CES 2014 might have begun by the time you are reading this article. One common theme that we can expect in a lot of products launched at CES this year would be the dependence on a multitude of sensors. These sensors are important for all the smart functionalities you see in high-end smartphones. Join us in this story as we check out the cutting-edge sensor technologies, and how you can implement them in your next electronic design.

Products released last year, and the Internet of Things itself, have acknowledged many sensor-based technologies. Some such technologies with the types of sensors used are discussed below. Anup Tapadia, founder, TouchMagix says, "The types of sensors that are emerging and have become more famous are the time-of-flight-based sensors that use reflected light pulses to measure the distance. These types of sensors have made 3D sensing technology famous."

## 9-axis sensor fusion technology.

This technology works with gyroscopes, accelerometers and magnetic sensors from any manufacturer without any restriction and thus allows us the required flexibility and control. Ideally, the precision of sensor fusion technology is very difficult and can delay the launch of a product as it includes so many steps, such as sensor integration, enhancement of signal, sensor drift, calibration, power consumption and magnetic drift. This technology uses low-power fusion co-processor with 9-axis sensor fusion algorithm that fuses and continuously auto-calibrates the sensors using a

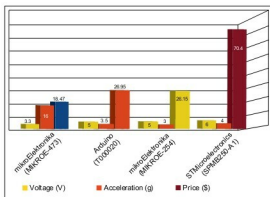


Fig. 1: Comparison of four accelerometer modules launched in the previous year

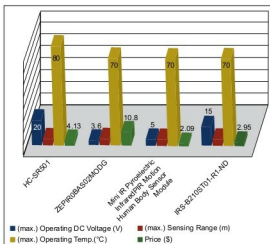


Fig. 2: Comparison of four motion sensor modules

small fraction (less than 1 per cent) of the power of even a low-power micro-controller sensor hub.

**SpacePoint technology.** It is also a 9-axis sensor technology in which 3-axis geomagnetic sensor is integrated with 3-axis accelerometer and 3-axis gyroscope, which is then fused with the 9-axis sensor fusion algorithm to get accurate motion sensing for consumer applications (like controllers for TV, set-top boxes, video games, etc). It provides a very accurate performance with high

resolution, effective responsiveness and 1:1 motion tracking. Imagine playing a video game and controlling the characters in the game just by your body actions with 360-degree pointing accuracy, even without touching the screen of the video game or any controller and, most importantly, without any time lag. This beautiful application has become possible due to SpacePoint technology.

**Electro-mechanical polymer (EMP) actuator technology.** EMP is a flexible, light and ultra-thin film with electro-active properties. It uses a piezoelectric polymer material which elongates on application of an electric field. This material is used to generate very fine and powerful sensors and actuators to enhance the sensory user experience in electronic devices. EMP, which is only about 200µ thick, stretches and elongates on application of a charge that creates vibration on the surface and causes sounds and surface deformation. The EMP touch sensors are considered perfect for the precise gesture-control applications such as squeezing detection and pressure. The sensors manufactured by this technology are ultra thin (<200µ), as light as a feather—weighing lesser than 0.001 gm and have a fast response time (<5 milliseconds).

Table I indicates different sensor-based technologies that have helped the launch of some amazing consumer